x11vnc Service - Installation Guide

Debian / Raspberry Pi Desktop for PC (Dell and other x86 computers)

OVERVIEW

install-x11vnc.sh installs x11vnc and xauth and configures
x11vnc-follow-session.service to start automatically at boot.
It shares the existing, active local X11 user desktop (seat0).
It does not create a new desktop or show the login screen.

REQUIREMENTS

- Debian-based PC using apt and systemd.
- Active local X11 graphical session; Wayland is not supported.
- Root/sudo permissions and internet access for packages.

1. INSTALL (DIRECT LAN ACCESS IS NOW THE DEFAULT)

```
sudo bash install-x11vnc.sh
```

The script requests a VNC password on first install, stores it in
/etc/x11vnc.pass, and enables the systemd service.
Classic VNC authentication uses at most eight password characters.

2. CONNECT USING THE DEVICE’S IP ADDRESS

On the Dell, find its address:

```
hostname -I
```

From VNC Viewer on another PC, enter:

```
DELL_IP
DELL_IP:5900
```

The server listens on TCP port 5900 while an active local X11
user session is available.

3. OPTIONAL: LOCALHOST-ONLY MODE WITH SSH TUNNEL

```
sudo bash install-x11vnc.sh --local
```

If SSH is enabled, on the remote Windows/Linux/Mac computer:

```
ssh -N -L 5900:127.0.0.1:5900 USER@DELL_IP
```

Connect VNC Viewer to 127.0.0.1:5900 over that tunnel.

SECURITY NOTICE

Direct VNC is not encrypted. Only use LAN mode on a trusted,
restricted network with port 5900 filtered by your firewall.
Never expose VNC directly to the public internet.

Operations and Troubleshooting

x11vnc-follow-session.service

4. CHECK OR MANAGE THE SERVICE

```
  sudo systemctl status x11vnc-follow-session
  sudo systemctl restart x11vnc-follow-session
  sudo systemctl enable --now x11vnc-follow-session
```

Read recent logs:

```
  sudo journalctl -u x11vnc-follow-session -n 40 --no-pager
```

Check the TCP listener:

```
  sudo ss -lntp | grep 5900
```

5. SWITCH BINDING MODE ON AN ALREADY INSTALLED PC

Edit /etc/default/x11vnc-follow-session and set:

```
  X11VNC_BIND=lan     # VNC Viewer connects directly to PC IP
  X11VNC_BIND=local   # requires SSH tunnel
```

Use only one of the settings above, then restart the service.
Updating a copy of the installer alone does not reconfigure
already installed machines.

6. NO VNC DESKTOP OR CONNECTION TIMEOUT?

- Confirm a user is logged into the local graphical desktop.
- In the graphical desktop terminal, check:

```
  echo "$XDG_SESSION_TYPE"
```

  This must show x11, not wayland.
- Verify the VNC service is active and read its logs.
- Check firewall access to TCP 5900 when using LAN mode.
- Check /etc/default/x11vnc-follow-session for the bind mode.

7. DEBIAN 11 (BULLSEYE) INSTALLATION SUPPORT

If normal APT operations fail with certain archive-related errors,
the installer can back up official Bullseye security sources,
switch them to Debian’s archived security mirror, and retry once.
This does not restore ongoing security support for Debian 11.

FILES INSTALLED

/etc/systemd/system/x11vnc-follow-session.service
/usr/local/sbin/x11vnc-follow-session
/etc/default/x11vnc-follow-session
/etc/x11vnc.pass

For details, see the repository README and shell script.